



Customer Shopping Behaviour Analysis

Uncovering actionable insights from ~3,900 purchase records across multiple product categories – spanning spending habits, segmentation, product preferences, and subscription trends.

Dataset at a Glance



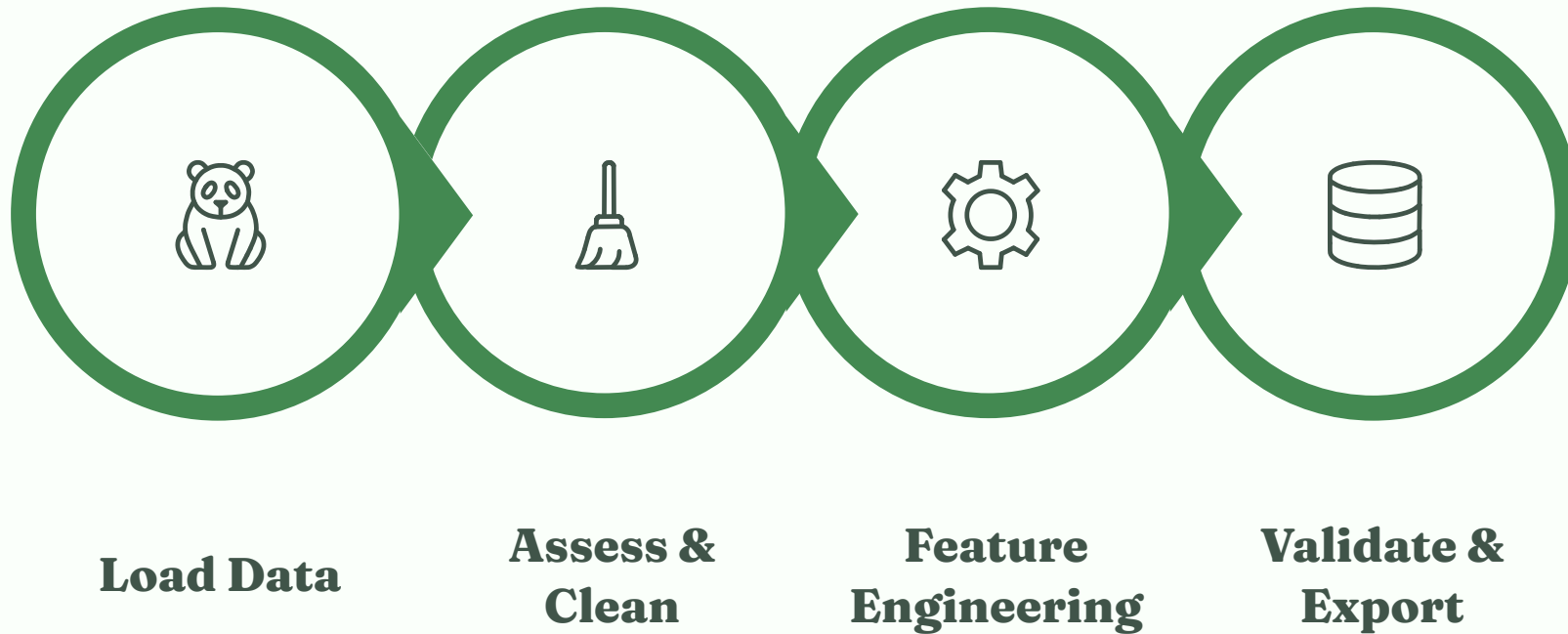
Dataset Specs

3,900 rows · 18 columns

- **Demographics:** Age, Gender, Location, Subscription Status
- **Transactions:** Product, Category, Purchase Amount, Season, Size, Colour
- **Behaviour:** Discount Usage, Purchase Frequency, Review Ratings, Shipping Preferences

❏ 37 missing values in Review Rating – imputed using median per product category.

Data Pre-processing Pipeline



Python was used end-to-end for data integrity – from loading and cleaning to feature creation and database export into MySQL for advanced querying.

Feature Engineering Highlights

Age Group Variable

Customers categorized into age brackets to enable demographic segmentation and age-based revenue analysis.

Purchase Frequency (Days)

Derived metric quantifying buying intervals – enabling classification of New, Returning, and Loyal customers.

Redundancy Removal

Identified overlap between `discount_applied` and `promo_code_used`; removed the less informative feature to reduce noise.

MySQL Business Analysis

Advanced queries extracted key insights across 10 analytical dimensions:



Revenue by Gender

Compared total revenue across male and female customers.



High-Value Discount Users

Identified customers using discounts while maintaining above-average spend.



Top-Rated Products

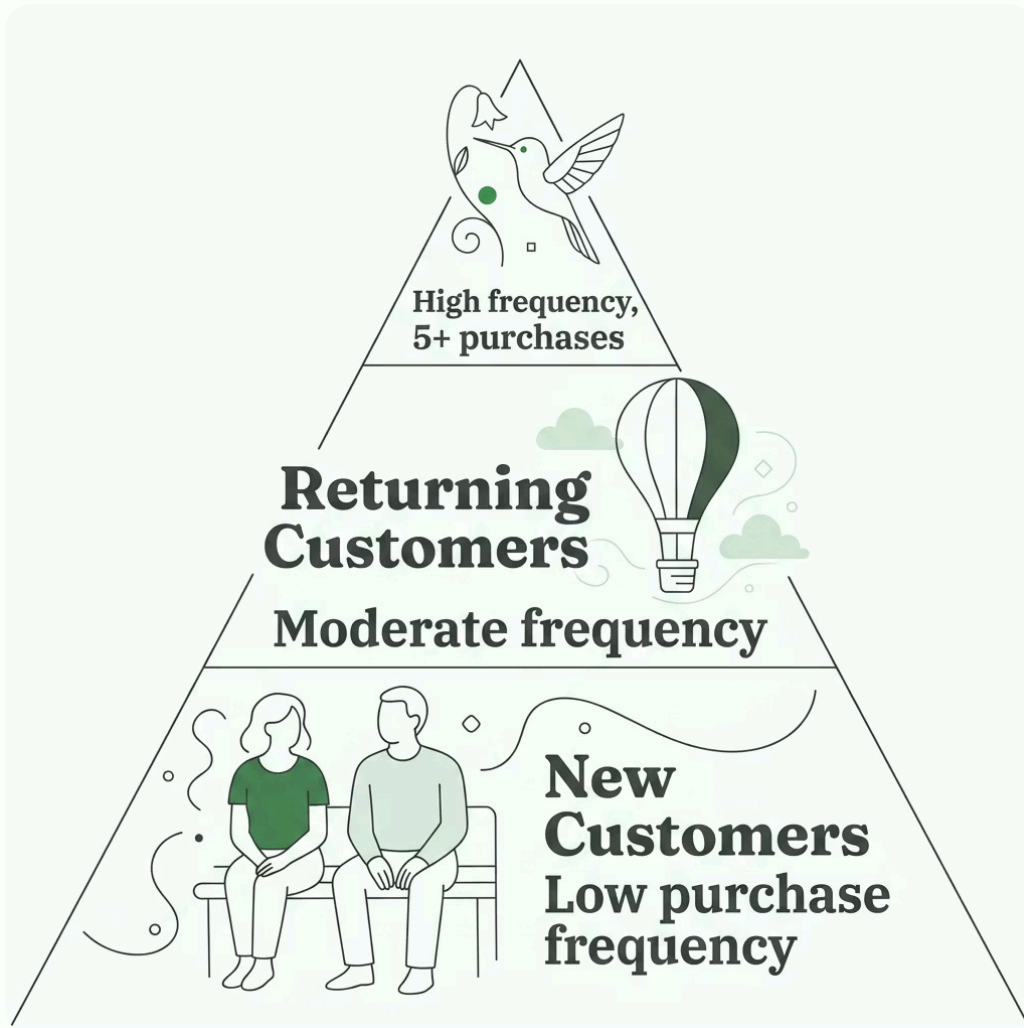
Ranked products by average customer review ratings.



Shipping Preferences

Evaluated purchase value differences between Standard and Express shipping.

Customer Segmentation & Subscription Insights



Segmentation Logic

Customers classified as **New**, **Returning**, or **Loyal** based on purchase frequency.

Subscription Impact

Subscribers demonstrated stronger spending patterns and higher revenue contribution versus non-subscribers. High-frequency buyers (>5 purchases) showed significantly greater subscription likelihood.

Product & Category Performance

Top-Rated Products

Ranked by average review rating – highlights quality leaders for promotional focus.

Category Leaders

Top-performing products identified within each category to guide inventory and marketing priorities.

Discount Sensitivity

Products most frequently purchased under discount conditions – informing smarter promotional targeting.



Power BI Dashboard



Interactive Dashboard Features

- Revenue trends and KPIs
- Customer segmentation visuals
- Product performance analysis
- Discount usage and shipping behaviour insights

✓ Built in Power BI for intuitive, stakeholder-ready reporting.

Revenue Insights: Age & Payment

Revenue by Age Segment

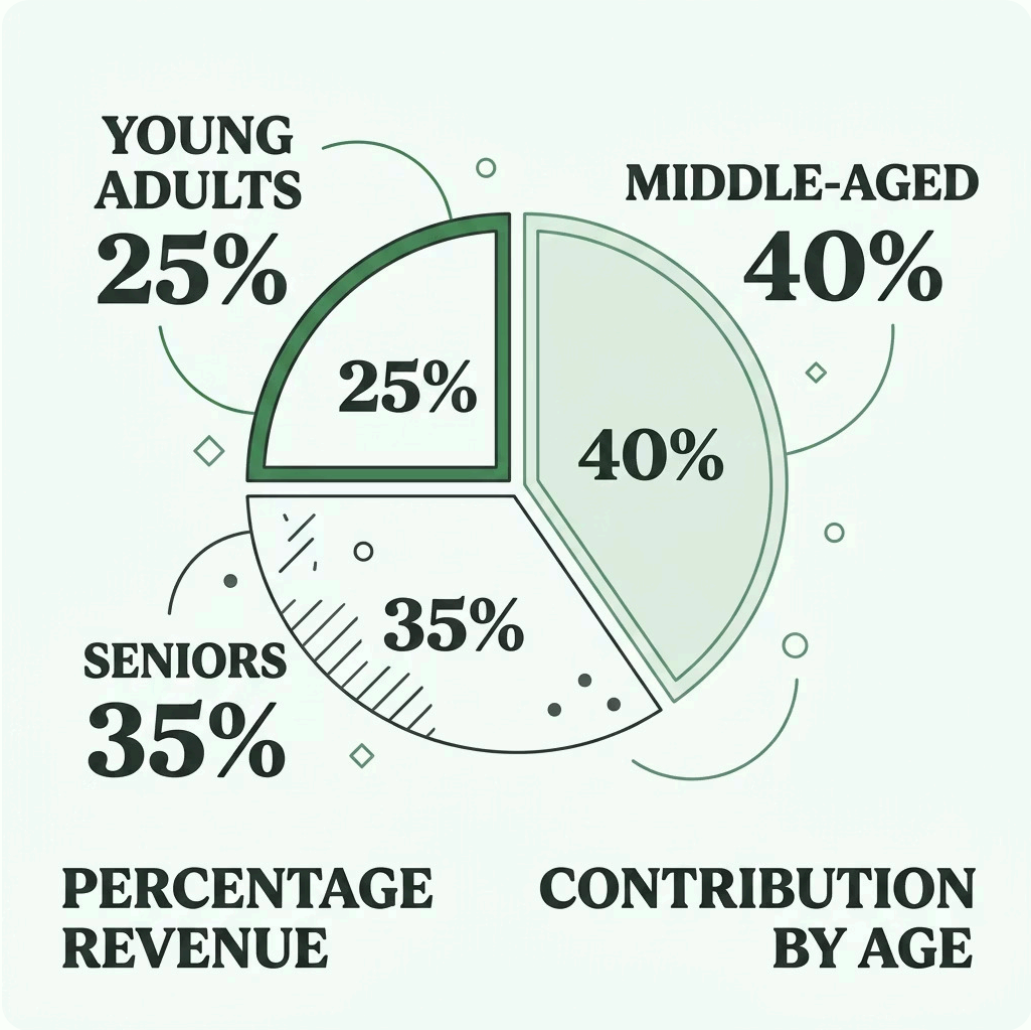
Different age groups contribute varying shares of overall revenue – enabling targeted demographic marketing.

Payment Method Usage

Analysis identified the most utilised payment methods, informing checkout experience optimisation.

Rating vs. Spending

Explored correlation between review ratings and purchase amounts to understand quality-driven spending behaviour.



Strategic Recommendations

→ Enhance Subscriptions

Introduce exclusive incentives to boost subscriber conversion rates.

→ Strengthen Retention

Implement loyalty programs to transition repeat customers into high-value segments.

→ Optimize Discounts

Balance promotions to maximize revenue without eroding profit margins.

→ Targeted Engagement

Focus marketing on high-revenue demographics and express delivery customers.

